Course Title	Electromagnetic Theory	**Course Code**	EE3005
Department	Department of Electrical Engineering (DEE)	**Campus**	Lahore
Knowledge Profile	Engineering Fundamentals (WK3)	**Credit Hrs.**	3
Knowledge Area	Electrical Engineering Fundamental (KA06)	**Grading Scheme**	To be announced by the Instructor
HEC Knowledge Area	Electrical Engineering Core (Breadth)	**Applicable From**	Fall 2022
Pre-requisite(s)	MT2008 Multivariable Calculus		

Course Objective	To be able to apply vector analysis to solve electrostatic and magneto static problems with an introduction to electromagnetic phenomenon governed by Maxwell's equations

No.	Assigned Program Learning Outcome (PLO)
1	An ability to apply knowledge of mathematics, science, engineering fundamentals and an engineering specialization to the solution of complex engineering problems
2	An ability to identify, formulate, research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences.
12	An ability to recognize importance of and pursue lifelong learning in the broader context of innovation and technological developments.

I = Introduction, R = Reinforcement, E = Evaluation, A = Assignment, Q = Quiz, M = Midterm, F=Final, L = Lab, P = Project, W = Written Report.

No.	Course Learning Outcome (CLO) Statements	Assessment Tools	Taxonomy Levels	PLO
1	Demonstrate the use of 3D orthogonal coordinate system and vector analysis tools (curl, divergence, etc.) in problem solving	Q1, A1, M1, F	C3	1.1L 2.3H
2	Formulate electrostatic fields and/or its properties governed by Coulomb's / Gauss's law for a given charge distribution in free space and / or dielectrics.	Q2, M1, M2, F	C5	1.1M 2.1H
3	Formulate the capacitance with one dimensional potential variation using direct integration	Q3, A2, M2, F	C5	1.1L
4	Formulate magnetic fields and its effects for a given distribution of moving charges using laws of magnetostatics to establish important properties of Maxwell's equations	A3, F	C5	1.1M 12.3M

Text Book(s)	**Title**	Engineering Electromagnetics, 8th Edition (International)
	Author	William H. Hayt, John A. Buck
	Publisher	McGraw Hill
Ref. Book(s)	**Title**	Introduction to Electromagnetic fields, 3rd Edition
	Author	Clayton R. Paul, Keith W. Whites, Syed A. Nasar
	Publisher	McGraw Hill
	Title	Field and Wave Electromagnetics, 2nd Edition
	Author	David K. Cheng
	Publisher	Pearson Education

Week	Course Contents/Topics	Chapter*	CLO*
1	Introduction, Applications, Review of Vector Analysis, Concept of Field, 3D orthogonal coordinate systems (rectangular Cartesian coordinates)	1.1 – 1.3 (1)	1
2	Circular, Cylindrical and Spherical coordinate system	1.8, 1.9 (2.2)	1
3	Electrostatics: Charge, Coulomb's Law, and Electric Field Intensity	2.1	2
4	E field due to different charge distributions	2.2 – 2.5	2
5	Electric Flux Density, Gauss's Law	3.1 – 3.3	2
6	Application of Gauss's Law and Divergence Theorem	3.4 – 3.6	1, 2
7	Potential field of a point charge and system of charges, Potential gradient	4.1 – 4.6	2
8	Energy density in electrostatic field, conductor properties and boundary conditions	4.8, 5.4	2
9	Dipoles, Method of Images, Dielectric Materials and boundary conditions for perfect dielectrics	4.7, 5.5, 5.7, 5.8	2
10	Capacitance, Laplace's equations with examples	6.1 – 6.3, 6.6, 6.7	3
11	Charges in motion, current density, Resistance, Ohm's law	5.1 – 5.3	1, 4
12	Magnetostatics: Biot-Savart Law, Ampere's Law	7.1, 7.2	4
13	Application of Ampere's Law, Curl and Stoke's Theorem, Magnetic flux and density with static Maxwell's equations	7.3 – 7.5	1, 4
14	Force on a moving charge, differential current element, and force between differential current element	8.1 – 8.3	4
15	Force and torque on a closed circuit, Inductance calculation	8.4 – 8.7, 8.10	4
16	Faraday's Law, Displacement current, Maxwell's equations	9.1 – 9.4	4

***Reference book chapters are given in brackets**

Assessment Tools	Weightage
Quizzes, Assignments	20%
Midterms (I+II)	30%
Final Exam	50%